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Learning to multitask: Effects of video game practice on electrophysiological indices of attention and resource allocation

EDWARD L. MACLIN,[a] KYLE E. MATHEWSON,[a,b] KATHY A. LOW,[a] WALTER R. BOOT,[c] ARTHUR F. KRAMER,[a,b] MONICA FABIANI,[a,b] AND GABRIELE GRATTON[a,b]

[a]Beckman Institute for Advanced Science and Technology, University of Illinois, Urbana, Illinois, USA
[b]Department of Psychology, University of Illinois at Urbana–Champaign, Urbana, Illinois, USA
[c]Department of Psychology, Florida State University, Tallahassee, Florida, USA

Abstract

Changes in attention allocation with complex task learning reflect processing automatization and more efficient control. We studied these changes using ERP and EEG spectral analyses in subjects playing Space Fortress, a complex video game comprising standard cognitive task components. We hypothesized that training would free up attentional resources for a secondary auditory oddball task. Both P3 and delta EEG showed a processing trade-off between game and oddball tasks, but only some game events showed reduced attention requirements with practice. Training magnified a transient increase in alpha power following both primary and secondary task events. This contrasted with alpha suppression observed when the oddball task was performed alone, suggesting that alpha may be related to attention switching. Hence, P3 and EEG spectral data are differentially sensitive to changes in attentional processing occurring with complex task training.

Descriptors: Event-related brain potential (ERP), Event-related spectral perturbation (ERSP), Electroencephalogram (EEG), Video games, Alpha rhythm, Attention control, Training, Practice

Video games have recently received increased interest from cognitive researchers, both as a model for the learning of complex tasks and as a means to examine training and transfer of skills (Achtman, Green, & Bavelier, 2008; Basak, Boot, Voss, & Kramer, 2008; Erickson et al., 2010; Green & Bavelier, 2008). As video games can approach real-life tasks in terms of complexity, they generate an ideal test bed for examining attention control functions when several tasks have to be performed concurrently. Here we exploit the features of a well-characterized video game (Space Fortress, SF; Donchin, Fabiani, & Sanders, 1989) and use electrophysiological measures to examine how attention to different tasks and task components is deployed before and after extensive training. This allows us to capture the dynamics of the attention control system as it changes with practice.

The SF game was developed as a tool for studying learning and training strategies (Donchin et al., 1989; Shebilske et al., 2005; see Figure 1). Players are required to focus attention on multiple demanding and overlapping component tasks that simulate the complexity of many important real-world tasks such as piloting, air traffic control, and radar or sonar monitoring. The game incorporates difficult motor, memory, dual-task, and visual/attentional components, many of which are based on the cognitive psychology literature. In short, it requires participants to control a flying ship and destroy an aggressive fortress while identifying mines that appear periodically as "friends" or "foes" and reacting to them accordingly.

In this article we focused on both discrete (event-related) and continuous aspects of the SF game. The discrete components include two nonflying aspects of the game: brain responses to successful "hits" on the fortress and to the appearance of mines. We selected these events for two main reasons: Their discrete nature allows for the computation of event-related activity, and they represent distinct types of events, in that mine appearance is independent of the participants' behavior, whereas fortress hits are a direct result of the participants' performance. To the extent that these two types of events engage different aspects of attention (reorienting of attention in the case of mine appearance and feedback processing in the case of fortress hits) we expect to observe different patterns of electrophysiological activity (Min & Hermann, 2007; Sauseng et al., 2005) and perhaps different patterns of change with game learning (Li, Gratton, Yao, & Knight, 2010; Zhuang et al., 1997).

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Address correspondence to: Gabriele Gratton, University of Illinois, Beckman Institute, 405 N. Mathews Ave., Urbana, IL 61801, USA. E-mail: grattong@illinois.edu

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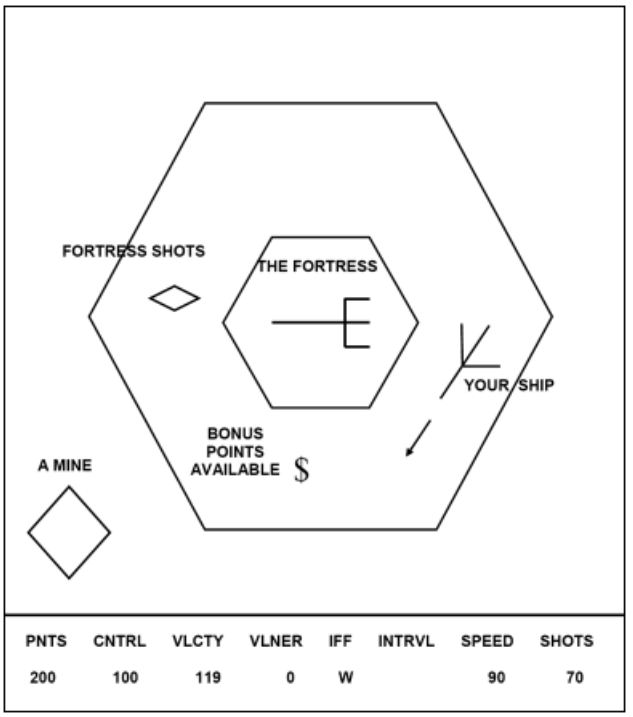


Figure 1. Schematic representation of the Space Fortress game. Subtask scores are continuously updated and displayed in the bottom panel.

In addition, we focused on a continuous component of the task, likely dominated by ship control, which needs to be carried on uninterruptedly. Note that this is the most critical aspect of game playing and has a large influence on game performance and learning (Fabiani et al., 1989). For this we analyze the background electroencephalographic (EEG) activity irrespective of discrete game events.

The data presented here were collected in the context of a larger study on the effects of training strategy on learning transfer (Boot et al., 2010) during which participants received 20 h of game training. The EEG recording sessions took place before and after training. During these sessions, a secondary probe task (auditory oddball) was introduced both during the game and in a separate condition where only the oddball task was performed. For this task, participants counted rare target tones in an ongoing sequence of frequent but irrelevant tones. The secondary oddball task was introduced as a means of tracking the change in allocation of resources that may accompany performance improvements in the primary task (SF game). With learning, resources originally dedicated to particular game events may be freed up and reallocated to processing the secondary task. Specifically, we were interested in comparing fortress hits and oddball tones because these two events share some common features. Namely, successful performance requires the participant to keep track of the number of occurrences of each: Fortress hits must be counted in order to know when the fortress has sustained enough damage to be destroyed with a fatal double shot, and a running count of the tones has to be kept to perform the oddball task.

To analyze the brain activity associated with these discrete task events we used two approaches: an analysis of event-related brain potentials (ERPs) and an analysis of event-related spectral perturbations (ERSPs). Note that the main difference between these two analytic approaches is that the ERPs assume coherence of the timing of brain activity to the eliciting events whereas the ERSPs do not (Makeig, Debener, Onton, & Delorme, 2004; Pfurtscheller & Lopes da Silva, 1999; Sauseng et al., 2007). Given the complexity of the game, we felt that it was important to examine both, in part because there could be increased variability of responses compared to standard laboratory tasks. In addition, ERPs and ERSPs may be differentially sensitive to neurophysiological transactions associated with cognitive processing.

For the ERP analyses we focused on the P300 or P3 component, which has been extensively used as an index of the allocation of attention across tasks (Allison & Polich, 2008; Isreal, Chesney, Wickens, & Donchin, 1980; Kramer, Wickens, & Donchin, 1983; Low, Leaver, Kramer, Fabiani, & Gratton, 2009; Sirevaag, Kramer, Coles, & Donchin, 1989). In a typical experiment, the difficulty of a primary task (or the priority given to that task) is manipulated while participants are asked to simultaneously perform a secondary task, such as an oddball task.[1] The amplitude of the P3 component to the secondary oddball task decreases when the primary task becomes harder or is prioritized, provided that the increased effort involves perceptual or central processing, or both, as opposed to strictly motor processing. We reasoned that with increased skill level on the game, attentional resources would be more effectively allocated and fewer resources would be required to play the game efficiently, and therefore more resources could be dedicated to the secondary task. By comparing the P3 elicited by the in-game task of hitting the fortress with that of the secondary task of oddball counting, we sought to quantify this trade-off in attentional resources.

Several studies have shown that some aspects of oddball ERSPs are associated with the P3 (Basar-Eroglu, Basar, Demiralp, & Schürmann, 1992; Bernat, Malone, Williams, Patrick, & Iacono, 2007), including power increases in the delta and theta bands and decreases in alpha power. Increases in delta and theta power and decreases in alpha-band power (alpha suppression, or event-related alpha desynchronization) have also been associated with the degree of attentional engagement (Klimesch, Doppelmayr, Russegger, Pachinger, & Schwaiger, 1998; Klimesch, Sauseng, & Hanslmayr, 2007). A series of recent studies has shown that attentional engagement can be tracked with some temporal precision by alpha oscillations (Babiloni et al., 2004). Other studies have demonstrated that these oscillations are directly related to the excitability of the cortex (Romei et al., 2008) and predict the conscious detection of visual stimuli (Mathewson, Gratton, Fabiani, Beck, & Ro, 2009). Here we espouse the view that the alpha rhythm occurs in many, perhaps all, cortical regions, and that its overarching role presumably reflects a pulsating inhibition process (Mathewson et al., 2009), although specifically adapted to each region's function(s) (Jensen & Mazaheri, 2010). In other words, we hypothesize that the role of alpha across regions is gating or controlling the flow and timing of information transfer, just as we have recently demonstrated in the visual system (Mathewson et al., 2009). Therefore we also measured induced changes in multiple frequency bands timelocked to the occurrence of significant events in the SF game and

[1]Here we assume that working memory and attentional limitations occur at a supramodal level, where a single resource pool is used to process visual and auditory stimuli (see Wickens, 2008). To use a dual task paradigm to study the allocation of this type of central processing resource (in contrast with modality-specific resources), the two tasks are typically presented in different sensory modalities and require different response modalities (counting and motor responses, respectively).

to the secondary task oddball tones, to further identify electrophysiological correlates of attentional engagement and how they change with training. Specifically, we predicted that the P3 and delta/theta power changes in response to fortress hits would decrease after training, whereas those elicited by rare tones would increase.

Past research has also shown a general increase in centrally located alpha and frontal theta activity with the learning of the SF task (Smith, McEvoy, & Gevins, 1999; see also Gevins, Smith, McEvoy, & Yu, 1997). Smith and colleagues also found that alpha recorded over somatosensory regions increased with acquisition of motor skill. We sought to replicate this finding by measuring alpha power in the period prior to discrete game events and extend it by examining the ERSPs elicited by such events.

Finally, activity in the beta range is also known to be important for attention and motor processing. Occipital beta activity has been shown to increase with attentional processing (Gross et al., 2004) and to have an inverse relationship with occipital alpha activity, such that lower posterior alpha activity is associated with higher beta activity. We thus predicted a generally complementary pattern of beta compared to alpha power.

Methods

Participants

Thirty-nine participants were recruited from the Urbana–Champaign community through flyers posted in campus buildings and businesses or through advertisements posted to online bulletin boards. Responding individuals completed a survey of their video game habits and returned it via e-mail. Participants were admitted into the study if they reported playing less than 3 h of video games a week in the past 2 years. All participants reported being right-handed and having normal or corrected-to-normal vision. Participants were paid $15 an hour for a total of $350. This compensation reflected the extensive time commitment required of them and provided a necessary incentive for them to complete all phases of this lengthy study. Two participants were excluded from data analysis, one because of problems with EEG recording artifacts and the other because of performance data well outside the group's distribution. Of the remaining 37 participants, 11 were male, and their ages ranged from 19 to 29 years. This study was approved by the University of Illinois Institutional Review Board.

Procedures

In an initial orientation session, participants watched a 20-min movie that explained all the details of the SF game, and then another 5-min movie that summarized the most important rules. After viewing these movies, participants played twenty-four 3-min-long games to familiarize themselves with SF and were then scheduled for their pretraining EEG session. After this, participants practiced the game for 20 h and then returned for their posttraining EEG recording session.[2] During the EEG acquisition sessions participants first performed three 70-tone blocks of the auditory oddball task without the game (oddball without game). They then played ten 3-min-long games and were told to try to maximize their total score in each game. While playing the 10 games, participants also performed a secondary auditory oddball task, also with 70 tones (oddball with game). For both oddball tasks, tone bursts were 340 ms long and were presented over speakers at ~70 dB SPL every 2330 ms (stimulus onset asynchrony). Frequent tones (~80%) were 350 Hz and rare tones (~20%) were 500 Hz. Participants were instructed to silently count the rare (high) tones and report the total at the end of each game. Silent counting is a common measure of attention in oddball tasks when, as in the present study, it is not possible (or desirable) to collect overt responses on each trial because they may interfere with the responses required by the primary task. The oddball task was implemented in E-Prime running on a separate computer.

The Space Fortress Game

Complete details of the SF game can be found in previous reports (Donchin et al., 1989), so here we summarize only the most important aspects of the game. Players navigate a ship in a frictionless environment with precise control using a joystick (see Figure 1). Players can rotate the ship by moving the joystick left or right and can apply thrust by pushing forward on the joystick. For players to slow or stop the ship they must rotate it so that it faces opposite the desired course correction and apply thrust. Control of the ship is a challenging and demanding task.

The player's main goal is to destroy the Space Fortress (at the center of the screen) as many times as possible while avoiding damage to his or her ship. To destroy the fortress, players must hit it with missiles by aiming their ship toward it and pressing the fire button on the top of the joystick. To destroy the fortress players must first hit it with 10 missiles, with the timing between each hit being at least 250 ms and then hit it again with a rapid ($<$250 ms) double shot. If the player hits the fortress with a double shot before it is vulnerable, the vulnerability of the fortress is reset to zero and the player must start over again accumulating hits. The fortress rotates and shoots back at the player's ship, so the player's ship must be in constant motion to avoid damaging hits. When the player's ship is hit four times it is destroyed and points are lost.

Mines appear on the screen at regular intervals, actively pursue the ship, and will damage the ship if they come in contact with it. Importantly, the fortress cannot be damaged or destroyed as long as a mine is on the screen; thus mines must be dealt with immediately. Responding to the appearance of a mine involves a Sternberg memory search task (Sternberg, 1966). At the beginning of each game participants are asked to remember three letters that identify foe mines; all other mines are friends. Each mine

[2]Participants were randomly assigned to receive either Full Emphasis Training (FET, $n = 19$) or Variable Priority Training (VPT, $n = 18$). Participants in both training groups were treated identically until the first training session, which followed the first EEG recording session. During training, FET participants were instructed to attempt to obtain the highest overall score. VPT participants were instructed to focus their attention on a subset of skills during game play on different blocks of trials. The effects of these training strategies on performance have already been reported (Boot et al., 2010). During the EEG recording sessions (pre- vs. posttraining), the group differences in performance were less pronounced (Group $\times$ Pre/post interaction: $F(1,36) = 3.28$, $p = .08$), possibly because all games were played in a dual-task context. Furthermore, training group did not interact with any of our electrophysiological measures. Therefore, all results reported here have been collapsed across training groups.

has a letter associated with it that is displayed in the instrument panel at the bottom of the screen, which identifies it as friend or foe (Figure 1, "IFF"). If the mine that appears is a friend, the player can shoot it, and the mine will transfer this shot to the fortress. However, if it is a foe, it must be flagged as such using the mouse and then destroyed with a shot. A friend mine misidentified as a foe becomes a foe mine that cannot be destroyed. Responding to a foe mine as if it were a friend results in wasted time as the identification process must be repeated. To succeed in the game, participants must be careful to remember which letters represent foe mines and to respond to mines appropriately.

Participants were comfortably seated in front of a 19-in. color LCD monitor and made game inputs using their left hand on the computer mouse and their right hand on a Logitech Attack 3 Joystick® for flying and shooting. No game-related sounds were presented to the subjects.

EEG Recording and Analysis

EEG data were collected in a sound- and electrically attenuated recording chamber from 64 electrodes[3] embedded in a flexible electrode cap (ElectroCap International, Inc.). Impedances were below 10 kΩ. Data were analog filtered online using a 0.01 to 30-Hz bandpass and sampled at 100 Hz. Note that game event timing accuracy was limited by the SF game refresh cycle (49 ms). This was not the case for the oddball tones, whose timing accuracy was limited only by the EEG sampling rate. The jitter introduced by this relatively coarse game event timing accuracy will strongly affect faster (>10 Hz) ERP components, whereas ERSP activity at these frequencies could still be observed. Electrooculogram (EOG) data were collected on bipolar channels from electrodes above and below the left eye (VEOG) and on the left and right outer canthi (HEOG). Scalp channels were referenced to the left mastoid during data collection and then rereferenced off-line to the average of the two mastoids. It should be noted that no attempt was made to formally localize any of the electrophysiological activity reported here. All references to anatomical locations reflect scalp topography.

EEG data were analyzed with custom scripts written using Matlab and EEGlab (Delorme & Makeig, 2004). The data were epoched around three events of interest: mine onsets, fortress hits, and rare oddball tones. Eye movements were corrected using the algorithm described by Gratton, Coles, and Donchin (1983). After eye movement correction, epochs with voltage changes exceeding 500 μV were discarded. If an electrode showed excessive noise (more than 30% of the trials with voltage changes greater than 500 μV), the data for that electrode were discarded and replaced by spatially interpolated data using EEGLab's spherical interpolation function. Data were then either averaged, for the ERP analyses, or submitted to a wavelet-based time–frequency (TF) analysis to derive the baseline EEG and ERSP measures.[4]

For the ERP analyses, we focused on the P3, because this component is an index of attention allocation and can be recorded to all three of our events of interest. Mean amplitude was measured at Fz and Pz over an interval between 300 and 500 ms following the fortress hits and rare tones and between 350 to 700 ms following the appearance of mines. All time windows were chosen based on data from electrode Pz, where the P3 is typically maximal. Different measurement intervals were used because the P3 was delayed in the mine subtask. We conducted a repeated-measures analysis of variance (ANOVA) on this activity with Training (pre/post) and Electrode (Fz/Pz) as factors.

Wavelet analysis of the epoched data was performed with the *newtimef( )* function of the EEGLab toolbox (Tallon-Baudry, Bertrand, Delpuech, & Pernier, 1996). To optimize the balance of time and frequency resolution, a complex group of Morelet tapered wavelets was computed, with one cycle at the lowest frequency (1.6 Hz, 625 ms) and increasing up to 19 cycles at the highest frequency (30 Hz, 633 ms). To increase frequency resolution, data windows were zero padded with a ratio of 2. This allowed for visualization of frequency information from 1.6 Hz up to 30 Hz, and from −645 ms to 1640 ms around the event of interest. As some of the strongest effects we observed occurred at the boundary of the alpha and beta bands, we selected regions of interest in time–frequency space visually rather than using predefined bands (Shackman, McMenamin, Maxwell, Greischar, & Davidson, 2010).[5] The time frequency ranges used were as follows: alpha, 10–15 Hz, 300–700 ms; beta, 15–30 Hz, 400–500 ms; delta, 0–5 Hz, 250–600 ms; and theta, 5–8 Hz, 400–500 ms.

We carried out a split-half cross-validation procedure to support our selection of these customized time frequency ranges. This procedure was based on splitting the subjects into two groups (odd and even) and performing a time–frequency analysis on each group. The results of the two analyses were very similar (indeed, virtually identical). This indicates that, if we had selected the time–frequency intervals on the basis of only half of the subjects (training set) and used them to analyze the other half (cross-validation set), we would have selected the same ranges independently of which group (odd or even) we would have chosen as training or cross-validation sets. This provides a cross-validation for our range selection. The results of this cross-validation procedure are too extensive to be presented in the current article, but will be made available upon request.

[3]The electrodes used were Fp1, Fp2, Fpz, Fz, FCz, Cz, CPz, Pz, POz, Oz, AF3, AF4, F1, F2, F3, F4, F5, F6, F7, F8, FC1, FC2, FC3, FC4, FC5, FC6, C1, C2, C3, C4, C5, C6, T7, T8, CP1, CP2, CP3, CP4, CP5, CP6, TP7, TP8, P1, P2, P3, P4, P5, P6, P7, P8, PO1, PO2, PO3, PO4, PO7, PO8, O1, O2, and M2.

[4]These criteria resulted in rejection of less than 5% of the trials, except for a few (∼6) noisy electrodes that were interpolated as described in the text. A few entire blocks were excluded because of technical problems. There were approximately 140 rare tones presented during the game; after artifact rejection, subjects averaged 139 trials pretraining and 138 trials posttraining. In the no-game condition, fewer than 1% of trials were excluded, leaving a mean of 41 trials out of 42. Out of the 120 mine appearances, an average of 3 were excluded pretraining and 8 posttraining. The number of fortress hits varied across subjects from 66 to 1,002 (mean 442) pretraining and from 64 to 2,807 (mean 1,369) posttraining.

[5]It should be noted that (a) no bands perfectly captured the observed effects across all conditions and regions, and (b) many effects straddled the borders of traditionally defined bands. Part of the problem is that time–frequency analysis inherently trades off between temporal and frequency resolution. As the time windows for analyses were set at less than a second to visualize the dynamics of brain processes associated with game play, the frequency resolution was several hertz. This led to results whose frequency band borders were uncertain and, perhaps, even slightly distorted (an effect at 10 Hz may be confused with an effect at 12 Hz and so on). For this reason sticking to traditionally defined borders did not capture the results accurately, and we were forced to adapt our band definition to the data. In choosing the band borders, our goal was to find bands that encompassed the most prominent effects across conditions as a way of simplifying comparisons between conditions. Clearly, this approach ignores subtle differences between conditions and regions, but this is part of the rich complexity of this data set that we intend to continue exploring in an ongoing follow-up study.

The time–frequency data were analyzed in two ways. First, the raw power data were compared between the pre- and posttraining sessions for each of the three events of interest. Differences with training were measured at each point in time–frequency space using uncorrected paired sample t tests. By averaging this measure over the period before the onset of the stimulus of interest (− 600 to 0 ms), we visualized changes with training in the brain activity related to the continuous aspects of the task, which may provide a measure of the activity changes due to flying. Changes in selected frequency bands were investigated using a three-way repeated-measures ANOVA including Event, Training (pre/post) and Electrode (Fz/Pz) as factors.

To measure the time course of changes in brain activity associated with discrete events and for a more appropriate comparison with the baseline-corrected ERP data, we obtained time–frequency plots for which the average prestimulus power spectrum was subtracted from the spectrum estimated at each time point, for each frequency. The resulting ERSP data reflect changes in brain activity elicited by the events of interest. Activity before and after training was compared using a paired t test at each point in time–frequency space at electrodes Fz and Pz. Differences due to training were identified at various frequencies and lags following the three events of interest. We conducted a repeated-measures ANOVA on these ranges with Training (pre/post) and Electrode (Fz/Pz) as factors. To visualize the spatial distribution of these effects we generated spline-interpolated topographic maps, based on the average activity for each channel in the related time-frequency windows.

Results

Behavioral Data

Performance on the SF game improved substantially with training. The average game score during the pretraining EEG session was − 92.2 ($SE = 259.5$) but after training it increased to 4227.4 ($SE = 285.1$), $F(1,36) = 338.0$, $p < .001$. Error rate in the oddball task was estimated as the absolute difference between reported and true counts scaled by the true counts. To derive accuracy, this value was subtracted from 1. For the oddball-with-game task, tone counting accuracy increased following training on the SF game (pre = 82%; post = 91%); $F(1,36) = 28.8$, $p < .001$. Performance on the oddball task without the game was very high both pre- and posttraining (97%) and did not change as a function of training on the game (or time), $F < 1$, likely because of ceiling effects.

Electrophysiological Data

ERP changes with training. The grand average ERP waveforms for fortress hits, oddball tones, and mines and corresponding topographic maps of the P3 are shown in Figure 2. For fortress hits and oddball tones, a parietally focused P3 (or P3 complex) can be seen peaking around 400 ms. However, the effect of training on this component was quite different for these two events. For fortress hits, the P3 decreased with training, $F(1,36) = 43.2$, $p < .001$, and was larger at Pz compared to Fz: Pre/post × Electrode interaction, $F(1,36) = 5.1$, $p < .05$; $F = 39.6$ and 27.9, Pz and Fz respectively, $ps < .001$. In contrast, for oddball tones during the game, the P3 increased with training, $F(1,36) = 30.2$, $p < .001$, and the training effect did not interact with electrode, $F < 1$.

For comparison purposes, Figure 3 shows that P3 amplitude was greatly reduced when rare tones were presented during the game as opposed to outside the game, although this difference was less pronounced after training. Rare tones presented during the game were also processed differently from the frequent tones, even prior to training on the game (i.e., the ERPs elicited by rare tones were more positive than those elicited by frequent tones).

The P3 to the mines peaked at around 500 ms and was larger at Fz compared to Pz, $F(1,36) = 3.3$, $p = .08$. There was a trend for the amplitude to increase after training, $F(1,36) = 2.7$, $p = .11$, and this effect appeared to be larger at Fz than Pz. The interaction of training and electrode was not reliable, $F(1,36) = 1.9$, $p = .17$. This suggests that mine appearances continue to require focused attention, even after extensive training.

ERSP changes with training. Plots of the ERSP responses from Fz and Pz for each of the event of interest before and after training are presented in Figure 4. Because we are primarily interested in changes following game training (i.e., post vs. pre), thresholded difference plots (post–pre) of the ERSPs time-locked to fortress hits, oddball tones, and mines can be seen in Figure 5. The ERSP difference plots (post–pre) show several large effects around 500 ms following each of these events. We conducted a repeated-measures ANOVA on this activity with training (Pre/post) and Electrode (Fz/Pz) as factors. The delta frequency band (1.6–5.0 Hz) showed a decrease in power after training for fortress hits, $F(1,36) = 21.2$, $p < .001$, which was more pronounced at Pz compared to Fz: Training × Electrode interaction, $F(1,36) = 5.0$, $p < .05$; Fz pre versus post, $F(1,36) = 6.9$, $p < .01$; Pz pre versus post, $F(1,36) = 36.2$, $p < .001$. This may reflect the large P3 elicited by these stimuli. For the oddball tones within the game, there was an increase in delta power after training, $F(1,36) = 3.7$, $p = .06$, with no interaction with electrode, $F < 1$. This pattern corresponds to the changes in P3 with training for these two events. Moreover, the change in delta power was correlated with the change in P3 across subjects for both fortress hits and oddball tones, $r(35) = .35$ and .44, respectively, $ps < .05$, consistent with the notion that these measures represent the same (or highly overlapping) functional processes (Bernat et al., 2007).

The alpha frequency band (10–15 Hz) showed an increase following training for both fortress hits, $F(1,36) = 24.5$, $p < .001$, and rare tones, although for tones the difference tended to have a more frontal distribution: Training × Electrode interaction, $F(1,36) = 3.1$, $p = .09$; Fz pre versus post, $F(1,36) = 2.9$, $p = .09$; Pz pre versus post, $F < 1$. Thus, alpha and delta showed differential sensitivity to training for these two events.

For mines, practice on the game produced an increase across a broad band of frequencies at Pz, with some frequency bands also showing an increase at Fz. We divided the activity into four bands, delta (1.6–5 Hz), theta (5–8 Hz), alpha (10–15 Hz), and beta (15–30 Hz), and for each of these bands we performed a repeated-measures ANOVA with two factors: Electrode (Fz/Pz) × Training (pre/post). The delta and alpha bands showed an increase in power following training, $F(1,36) = 26.9$ and 7.0, respectively, $ps < .01$, but neither band interacted with electrode. In contrast, the theta and beta bands showed both an increase in power with training, $F(1,36) = 11.7$ and 8.2, respectively, $ps < .01$, and an interaction with electrode, $F(1,36) = 5.4$ and 11.8, respectively, $ps < .05$. For both of these bands, the power increase was greater at Pz than at Fz.[6]

[6]It may be possible that eye movements were not perfectly corrected and that there might have been systematic differences in eye movements from pre- to posttraining. This might account for the significant far

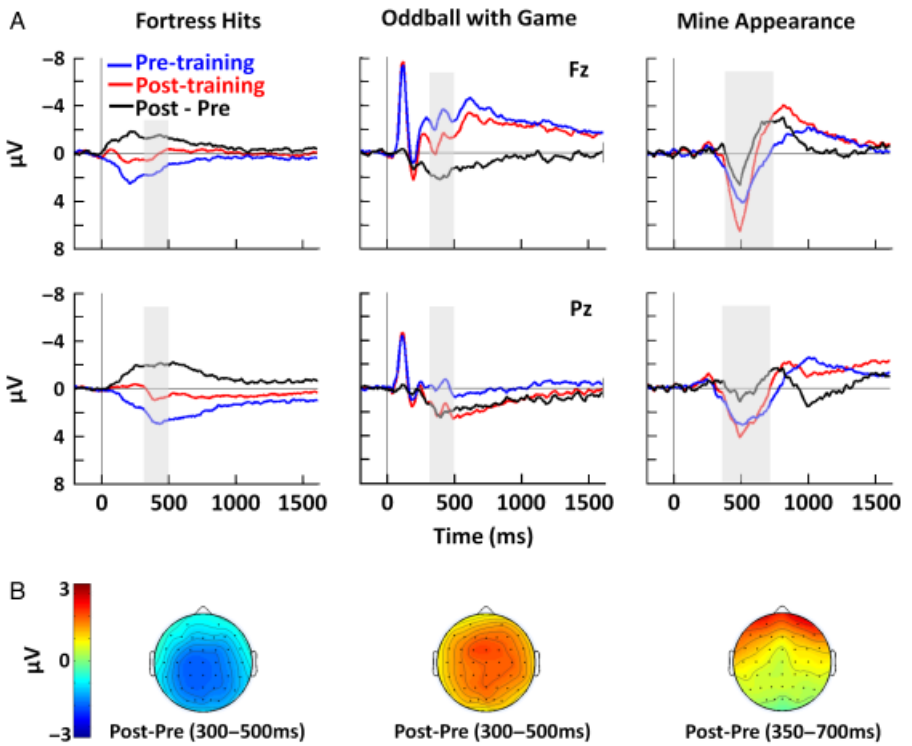


Figure 2. A: Grand-average ERP waveforms from Fz (top row) and Pz (bottom row) time-locked to SF game events. The black lines refer to the difference between the ERPs recorded before and after training (post–pre). The gray rectangles indicate the measurement intervals for P3. B: Voltage maps representing the change in scalp distribution of P3 between the post- and pretraining sessions.

Resource reciprocity between fortress hits and oddball tones. Both fortress hits and oddball tones required participants to keep track of the number of occurrences of each. Therefore, we were interested in directly comparing these two events. As can be seen in Figure 6, for both P3 amplitude and the delta band ERSP, there was a reliable crossover interaction of task (fortress hits vs. tones) and training, $F(1,36) = 53.4$ and 16.2, respectively, $ps < .001$. Moreover, across subjects, the training-related changes in processing of the fortress hits were negatively correlated with the training-related changes in processing of the oddball tones: delta, $r(35) = -.46$, $p < .01$; P3, $r(35) = -.20$, $p = .12$. Finally, the posttraining *decrement* in P3 amplitude to hits and the posttraining *increase* in P3 amplitude to tones were correlated with improvement in SF game performance: P3 for hits, $r(35) = -.28$, $p < .05$; P3 for oddball, $r(35) = .35$, $p < .05$]. Similar but smaller effects were observed for delta power: for hits, $r(35) = -.18$, $p = .14$; delta for oddball, $r(35) = .23$, $p = .09$. Thus, these electrophysiological measures provide a quantitative estimate of resource reciprocity, where resources that are originally required to process game events can be freed up after training and be reallocated to processing secondary task events without affecting primary task performance (Kramer et al., 1983; Wickens, Kramer, Vanasse, & Donchin, 1983). Although the probability of an oddball tone remained constant for the pre- and posttraining sessions, the frequency of hitting the fortress increased after training on the game. Therefore, the change in P3 amplitude to the fortress hits could be because of changes in frequency, but as you get more successful at hitting the fortress, presumably the processing related to those hits would also become less informative (or less effortful).[7]

Event-related alpha changes. Figure 4 shows a relatively brief change in alpha approximately 500 ms following event onset for most conditions, even before any practice with the game. Surprisingly, we did not see the expected suppression of alpha for events occurring during game play, but instead we found a transient increase for both mines and oddball tones, $F(1,36) = 20.8$ and 9.4, respectively, $ps < .01$ (see Figure 7 for a plot of post-stimulus alpha changes in response to different events). This is in sharp contrast to the expected decrease in alpha at Pz, which also occurred around 500 ms following rare tones when the oddball task was performed alone, $F(1,36) = 8.9$, $p < .005$. Although we chose a fixed frequency band (10–15 Hz) for all events, Figure 4 demonstrates that the alpha suppression seen after oddball tones without the game was even more robust at lower frequencies (8–11 Hz). Both the mines and the oddball tones have a working memory component, and previous research has shown increases in alpha with higher memory loads (Palva & Palva, 2007; Sauseng et al., 2005), so this finding may be related to some aspect of memory processing (assuming that memory load is higher in the

frontal effects apparent in some of the topographic plots in Figure 5. However, it is clear that both theta and beta (and to some extent alpha as well) distributions show clear maxima in posterior areas (around Pz), which cannot be because of eye movements. For this reason the results presented in this article cannot be completely accounted for by eye-movement effects.

[7]To further address the issue of the relationship between ERP and ERSP activity, we carried out a time–frequency analysis on the average (ERP) waveforms. These data indicate that most of the low-frequency ERSP activity (< 5 Hz) is because of ERP activity, but that the ERPs do not contain high frequency (> 5 Hz) activity. This suggests that high-frequency ERSP activity is independent of the dominant ERP activity present in the data. These analyses will be made available upon request.

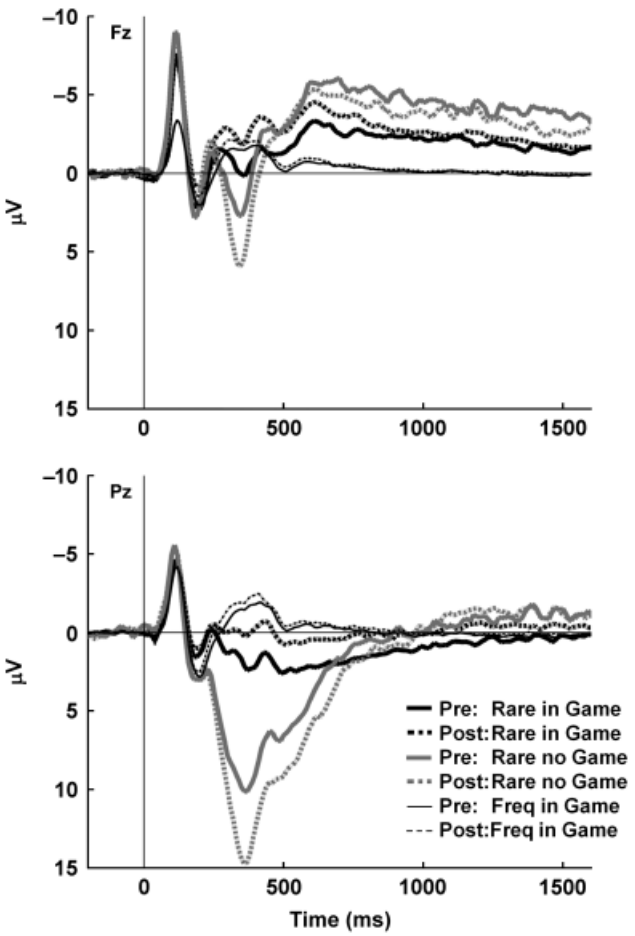


Figure 3. Grand-average ERP waveforms at Fz and Pz for rare and frequent tones during and outside the game, before and after training.

oddball with the game compared to the oddball without the game). Another possibility is that this transient increase in alpha may be related to the attentional switching inherent to this complex game environment. The lack of alpha increase to fortress hits is consistent with this latter interpretation because presumably gamers are tracking the shot after it is fired, so there would be no need to switch attention to a fortress hit.

Non-event-related EEG changes with training. Power spectrum changes due to training (post–pre) prior to fortress hits, oddball tones, and mine appearances can be seen in Figure 8. For this analysis, we focused on the 600 ms *preceding* the event onset, which correspond to periods dominated by maintenance of ship flight. The frequency distribution of post–pre changes was different in the baseline period compared to the event-related changes, so slightly different bands were used (alpha: 8–12 Hz, beta 25–30 Hz). In all three preevent periods there was an increase in alpha band power after training, $F(1,36) = 8.7$ and 9.7, $ps < .01$, prior to hits and tones and $F(1,36) = 2.7$, $p = .10$ prior to mines, and a decrease in beta band power after training, $F(1,36) = 4.8$, 3.8, and 4.8, prior to hits, tones, and mines, respectively, $ps < .05$, as predicted. A three-way repeated-measures ANOVA of Event × Training (pre/post) × Electrode (Fz/Pz) indicated that these baseline measures did not vary across the three conditions. For both the alpha and beta bands, there were no main effects of event type, $F < 1$. For the alpha band, there was a marginal Event × Training interaction, $F(1,36) = 2.9$, $p = .06$, such that the training-related increase was less pronounced prior to mines compared to the fortress hits and oddball tones. The baseline beta training changes were equivalent across the three events: Event × Training interaction, $F(1,36) < 1$. However, the decrease in beta power with training was more pronounced at Pz than at Fz: Training × Electrode interaction, $F(1,36) = 3.47$, $p < .07$; Fz pre vs. post, $F(1,36) = 1.9$, $p = .17$; Pz pre vs. post, $F(1,36) = 7.5$, $p < .01$.

Discussion

Significant improvements in performance were found with training for both the SF game and the secondary oddball task. Results demonstrated that electrophysiological measures (ERPs and ERSPs) can be used to analyze the participants' workload during a complex task (Craven, Tremoulet, Barton, Tourville, & Dahan-Marks, 2009; Rabbi et al., 2009), as well as the degree to which attentional resources are freed from the primary task and become available for secondary task performance.

Participants appeared to be able to allocate additional attentional resources to the secondary task as their skill in the primary task improved. This presumably reflects the automatization of game subtasks (Kramer, Fisk, Schneider, & Donchin, 1986; Kramer & Strayer, 1988). The present results illustrate physiological concomitants (P3 amplitude and delta power) of the allocation of shared resources (Allison & Polich, 2008; Isreal et al., 1980; Low et al., 2009; Sirevaag et al., 1989; Wickens et al., 1983) with practice in a complex game environment and also provide some level of specificity about which subtasks are less likely to become automatized, as demonstrated by the persistence and/or increase after training of brain activity following mine onset. In other words, the data are clearly consistent with the idea that some elements of the game (such as fortress hits) do change in salience with training (and thus elicit smaller brain responses) whereas others (such as mine appearance) do not. The inability of the subjects to develop automaticity for the mine identification task is congruent with previous work by Schneider and Shiffrin (1977), which shows that automaticity can only develop under consistent-mapping conditions (i.e., when the same memory set is used across the entire training). In the SF task, instead, the memory set is varied across blocks, generating a variable-mapping condition, which, according to Schneider and Shiffrin, does not allow for the development of automaticity. Another possibility is that complex game training may also increase overall resource capacity, benefiting secondary task performance without compromising even highly demanding primary tasks. The topography of brain activity following mine onsets showed a greater frontal contribution compared to that of fortress hits. The additional frontal activity may be related to the higher level of control processes demanded by the mine handling subtask. An alternative interpretation is that this frontal activity may be the result of systematic changes in eye-movement strategies to mine appearance occurring with training. However, it is important to note that eye movements are an important component of attentional shifts, and therefore even this interpretation can be considered consistent with the idea that attention control changes with training. Future studies of these physiological effects may shed light on just how resources are managed in a dynamic workload environment.

A morphological difference that can be noted in the ERPs elicited by mines, oddball tones, and fortress hits is that the P3s to

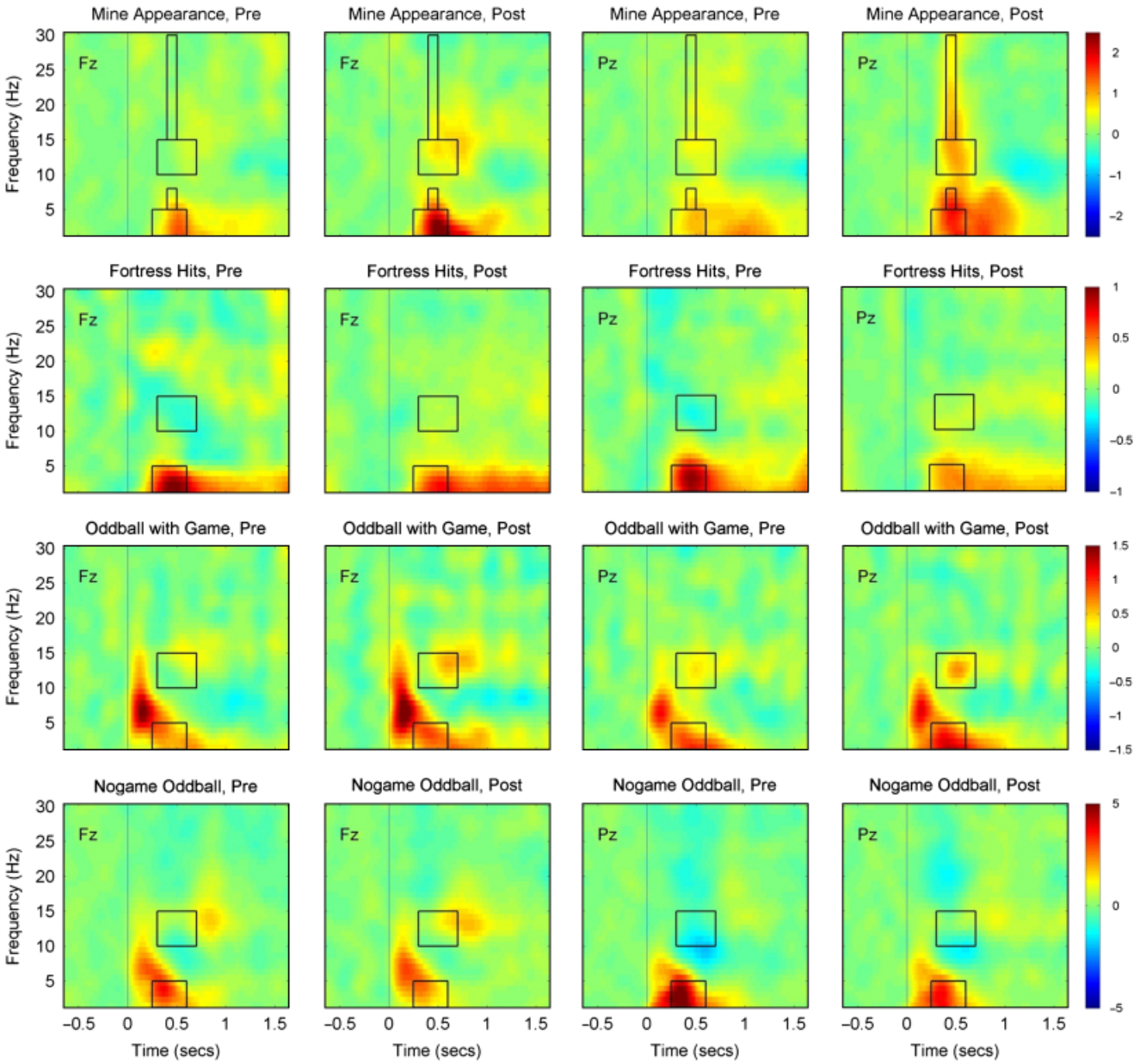


Figure 4. Event related spectral perturbation (ERSP) plots for fortress hits, rare tones (inside and outside the game), and mine appearances, before (pre) and after (post) training for electrodes Fz and Pz.

mines appear sharper, suggesting a decreased latency jitter. This may correspond to a greater degree of "cognitive" jitter of the subjects' reactions to the tones and fortress hits than to the mines, reflecting the fact that the mines require an immediate response, whereas the other two events do not. An alternative, and not mutually exclusive, explanation is that mine appearance also elicits a long-latency negative component that may diminish the late part of P3 amplitude because of overlap.

Several authors have proposed that the P3 reflects phase-locked delta, theta, and alpha range oscillations (Basar-Eroglu et al., 1992; Bernat et al., 2007; Yordanova, Kolev, & Polich, 2001). It is difficult to make definitive generalizations about the relative sensitivity of ERPs and ERSPs to specific aspects of cognition. However, our observation of an inversion of the alpha suppression normally found in single-task oddball experiments coupled with the differential effects of training (P3 decreases and alpha increases) clearly shows that these measures reflect different aspects of the underlying cognitive processing. One possibility is that the alpha enhancement we observe is related to attention-set switching, which would be greater during the game than in the oddball-alone condition. This is because the task is complex and requires paying attention to different elements at different times. The process of switching attention set may require both activation (of the processors related to the newly relevant elements) and inhibition (of the processors related to old, not currently relevant, elements). Thus, this account is not necessarily inconsistent with the general idea that alpha may be related to inhibitory processing. However, in this view, frontal alpha may be an expression of top-down cognitive control operations.

Specific high-temporal-resolution event-related effects were observed in multiple bands in the ERSP. These effects varied across event types, although in all cases the most prominent ones

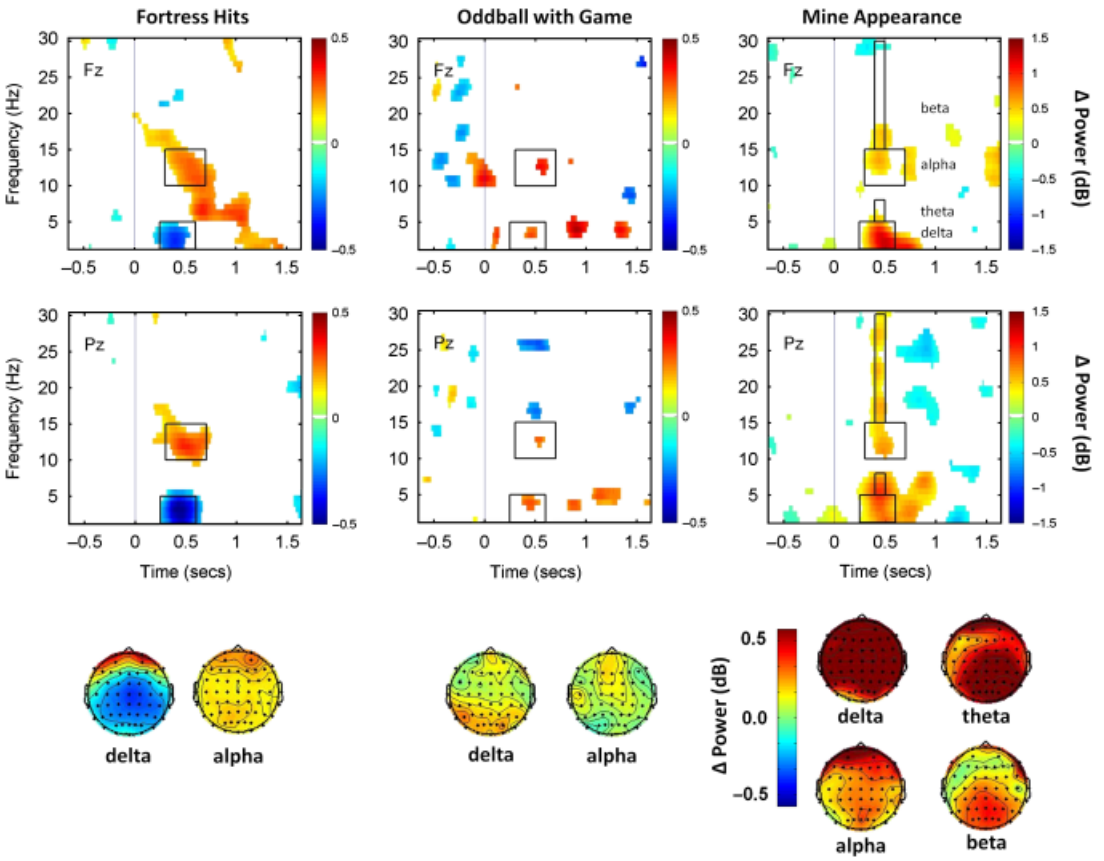


Figure 5. Top rows: ERSP time–frequency plots (first row, Fz; second row, Pz) of the differences in brain activity with training (post–pre). The plots only show differences that passed an uncorrected paired t test at each point in time–frequency space ($p < .05$). Boxes represent the time–frequency windows used for analyses. The baseline in the figures starts at the midpoint of the longest wavelet used for estimating the baseline correction spectra. Bottom row: Topographic plots of the power differences between after and before training.

occurred approximately 500 ms after the event. For fortress hits, the low-frequency band (1.6–5.0 Hz) showed a decrease in power after training, whereas the alpha band (10–15 Hz) showed an increase following training. For mine hits, training produced robust broadband (delta, theta, alpha, and beta) increases in activity. Training led to modest increases in the response to rare tones presented during the game in both the delta (1.6–5.0 Hz) and alpha (10–15 Hz) frequency bands.

It should be noted that in this study we made no attempt to control for effects of the relative overlapping of temporal sequences of events in different subtasks. This is an inherent characteristic of real-world multitasking and presents particular problems of experimental design when the sequences depend on the subjects' responses. In the future, both improved experimental designs and analysis methods are likely to provide further insights into how the brain adapts to simultaneous task demands.

Our results confirm previous work of Smith and colleagues (1999; see also Gevins et al., 1997) in showing a general (non-event-related) increase in alpha power with training during the SF session as a whole. Also consistent with this previous work,

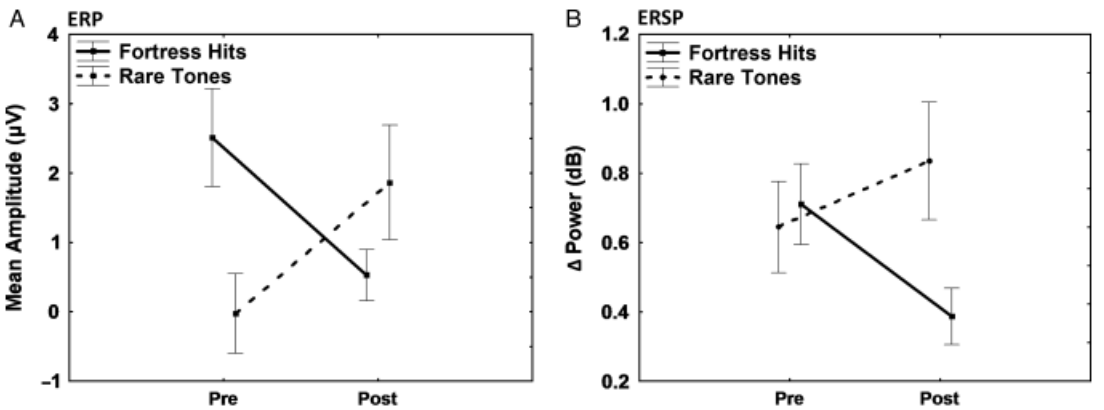


Figure 6. A: Mean amplitude of P3 at Pz before and after training for fortress hits and rare tones. B: Mean delta band ERSP power change at Pz before and after training for fortress hits and rare tones. Error bars in both plots indicate the standard error of the mean.

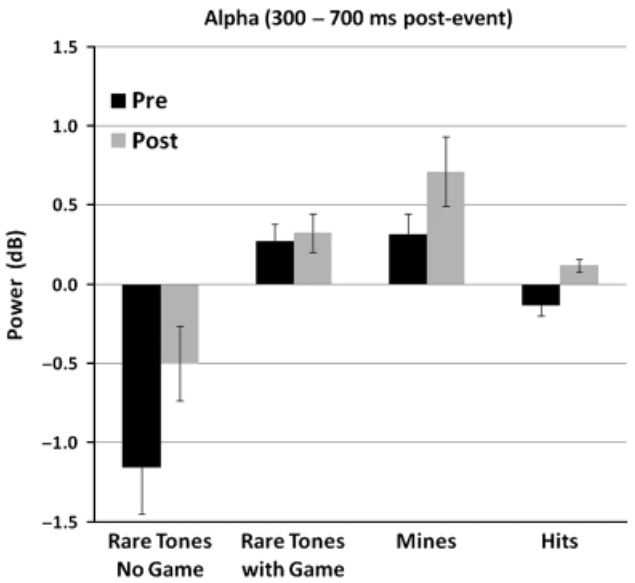


Figure 7. The mean ERSP power changes in the alpha band for each game event before and after training, between 300 and 700 ms after event onset. Error bars indicate the standard error of the mean.

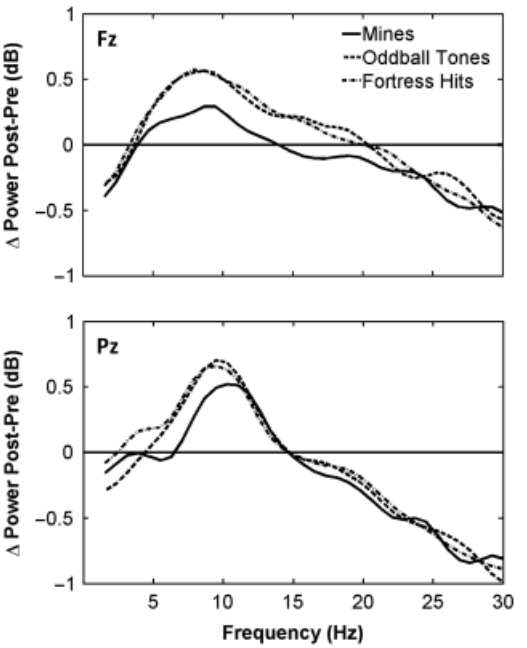


Figure 8. Power spectra differences during the periods preceding each discrete event (top, Fz; bottom, Pz).

our results indicate a relationship between alpha effects and those occurring at lower frequencies (delta and theta bands). This low-frequency ERSP activity largely overlaps with the ERP P3 effects and may reflect similar brain phenomena. Our findings extend the results presented by Smith and colleagues by demonstrating a decrease in beta power with training. We attribute this change to the learning of ship control skills.

The differences we observed in peak frequency, particularly in the alpha band, across conditions and electrodes suggest that there are multiple generators of activity in each band. We suggest, however, that the role of the activity in specific bands is similar in different regions of the brain and, in particular, that alpha activity reflects the control and gating of attention that is part of the process by which networks of regions are brought to bear on particular tasks (Jensen & Mazaheri, 2010; Mathewson et al., 2009).

In conclusion, the current study demonstrates that electrophysiological measures provide a window on the richness and complexity of brain activity associated with specific subtasks in a video game environment. Electrophysiological data are particularly useful because they can be used to monitor both continuous and discrete (event-related) components of the task. The use of a dual-task methodology further enhances our understanding of training effects. Future research may address issues of the interaction with training regimes and individual differences in learning abilities, strategies, and skill transfer.

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